Solution Architecture

Date	15 March 2026
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. The diagram maps the Presentation Layer, Application Layer, and Data Layer used in the VBCUA project.

Presentation / Client Layer
Streamlit Web App - concept input, audio upload, result dashboard, PDF download

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Application Controller / Streamlit Runtime
Handles routing, file upload, user inputs, and calls project modules

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Auth Service	Core Logic Service	External API Integrations
Not used currently single-user academic app	*Whisper, Sentence-BERT, Librosa, scoring engine*	*No paid external API; local/open-source models*

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Data / Storage Layer
No database - local runtime folders: data/uploads, data/waveforms, data/reports

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web interface where users enter the concept name, reference explanation, upload audio, view analysis results, waveform, and download PDF report.	Streamlit, Python
Application Controller	Manages user actions, saves uploaded audio, calls analysis modules, and displays final evaluation in the dashboard.	Streamlit Runtime, Python
Core Logic Service	Performs speech-to-text transcription, semantic similarity analysis, audio feature extraction, filler word detection, final scoring, and PDF generation.	OpenAI Whisper, Sentence-BERT, Librosa, SoundFile, NumPy, Matplotlib, ReportLab
Data / Storage Layer	Stores uploaded audio files, generated waveform images, and PDF reports temporarily during application runtime. No permanent database is used.	Local File Storage: data/uploads, data/waveforms, data/reports